



BITS Pilani

Automotive Vehicle

01/08/2026

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Department of Mechanical Engineering



Automotive Vehicle (AEL ZC441)

Lecture: Powertrain Fundamentals

Course Overview



- **Course Title**

Automotive Vehicles (AEL ZC441) - 3 Credit Units, 32-hour contact model.

- **Course Description**

Internal combustion engines; vehicle performance; analysis and design of vehicle components. Experimental or theoretical investigation of problems selected from the field of automotive vehicles.

- **Course Objectives**

Study of main automobile systems (engine, transmission, drive-axle, suspension, brakes); understanding working principles; performance analysis; and exposure to advanced concepts through projects and assignments.

How We Will Proceed



- **Format**

16 contact sessions delivered over the semester, each built around one focused topic from the Learning Plan.

- **Self-study expectation**

Students are responsible for maintaining the self-study schedule alongside contact sessions, per the course handout.

- **eLearn Portal**

All announcements, deadlines, and session material are posted on the BITS eLearn portal - check it regularly.

- **Assessment rhythm**

Continuous evaluation through quizzes/assignments (EC1), a Mid-Semester Test (EC2) after Session 8, and a Comprehensive Exam (EC3) covering all topics at the end.

- **Given this cohort's experience**

Sessions will move quickly through foundational material and spend more time on industry practice, trade-offs, and current developments.

Evaluation Scheme



Component	Type	Weight
EC1 - Quiz	Online	10%
EC1 - Assignment	Online	20%
EC2 - Mid-Semester Test	Closed Book	30%
EC3 - Comprehensive Exam	Open Book	40%

16 Session Learning Plan (Sessions 1-8)

Session	Topic
1	Introduction to Automobiles - classification, homologation, development methodology, virtual engineering
2	Chassis and Body - types of chassis and applications, weight distribution and analysis
3	Powertrain Fundamentals - propulsion requirements, tractive force and torque, powertrain architecture
4, 5, 6	IC Engine Fundamentals - engine basics, fuel conversion, torque characteristics, emissions
7	Electric Drive & Alternate Fuel Vehicles - BEV vs HEV, alternate fuel, fuel cell/hydrogen vehicles
8	Transmission Systems - manual, epicyclic, CVT, dual-clutch, AMT gearboxes

16 Session Learning Plan (Sessions 9-16)

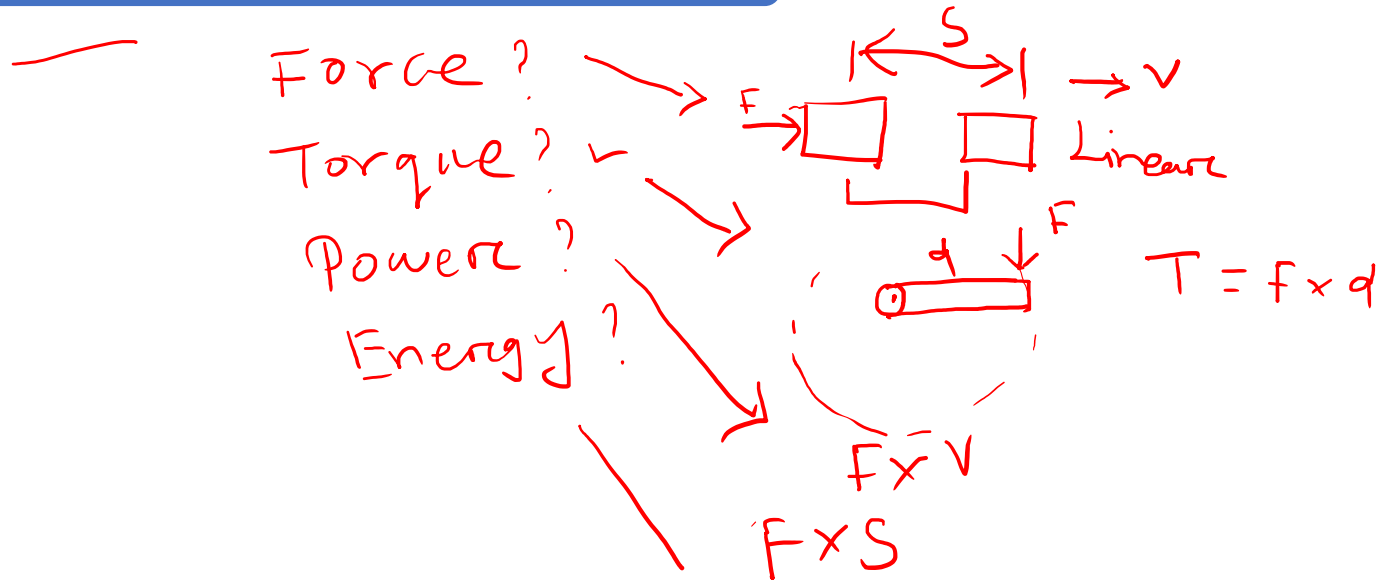
Session	Topic
9	Vehicle Control Fundamentals - E/E architecture, sensors and actuators, ECU architecture
10	Cooling & Lubrication Systems - heat generation in ICE and EV, lubrication
11	Vehicle Dynamics - fundamentals, influencing factors, numerical problems
12	Suspension - fundamentals, types, impact on dynamics
13	Steering Systems - fundamentals, dynamics, EPAS and SBW systems
14	Brakes - fundamentals, impact on dynamics, vacuum assist systems
15	Passive Safety - airbags, seatbelts, ABS systems
16	Active Safety Systems - ADAS and driver assist systems

Pre-requisite for the course: AEL ZC441



1. Basic Mathematics

2. Basic Physics



What Is Propulsion System?



The propulsion system is everything that generates and delivers the force needed to move an object.

Propel

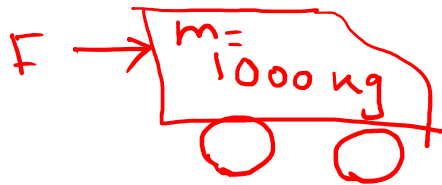


Why Propulsion System Needed?



Force and torque required to propel a car?

Target: A 1000 kg car to reach from 0 to 100 kmph in 5 second



0 $\rightarrow$ 100 km/h
in 5 Sec

$$u = 0, V = 100 \frac{\text{km}}{\text{h}} \rightarrow 100 \times \frac{1000}{3600} = \underline{27.8 \text{ m/s}}$$

$$V = u + a \times t$$

$$27.8 = 0 + a \times 5$$

$$a = \underline{5.56 \text{ m/s}^2}$$

$$g' = \frac{5.56}{9.81} = \boxed{0.56} \rightarrow$$

0.3 $\rightarrow$ Car
1 $\rightarrow$ Sports Cars

$$F = m \times a$$

$$= 1000 \times 5.56$$

$$= 5560 \text{ N}$$

Each person $\rightarrow F = 10 \text{ kg} \times 9.81$

$$= 98.1 \text{ N}$$

$$\approx 100 \text{ N}$$

No. of Person

$$\frac{5560}{100}$$

$$\underline{55.6}$$

≈ 56 ?

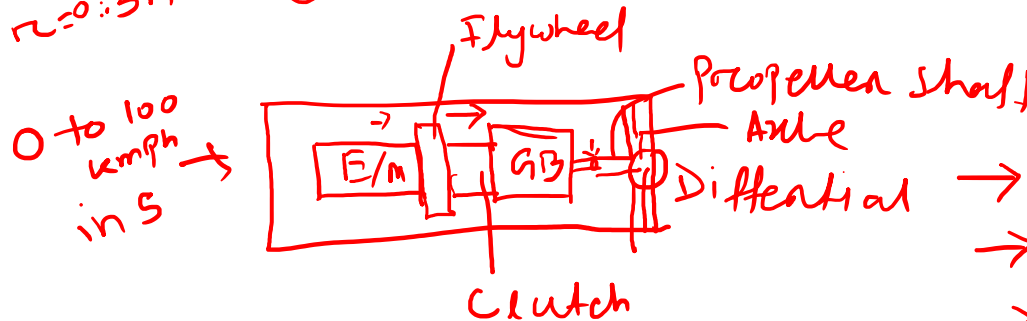
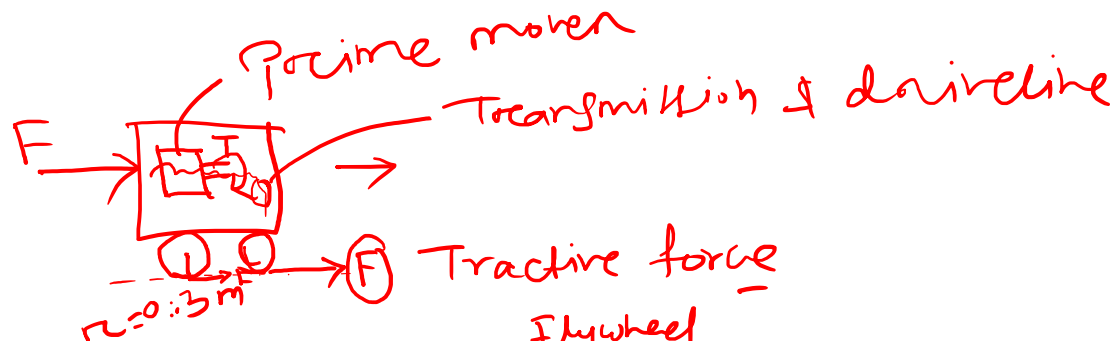
Why Propulsion System Needed?



Force and torque required to propel a car?

Target: A 1000 kg car to reach from 0 to 100 kmph in 5 second

Propulsion System → 56 persons job



$$F = 5560 \text{ N}$$

$$T = F \times r$$

$$= 5560 \times 0.3$$

$$T = 1668 \text{ Nm}$$

→ Rolling resistance
→ Air resistance
→ grade resistance

$$T = 2000 \text{ Nm}$$

Force and torque required to propel a car



A vehicle accelerates from 0 to 100 km/h in 5 s. Design a Propulsion system.



Given

$$\begin{aligned} m &= 1000 \text{ kg} \\ u &= 0 \text{ m/s} \\ v &= 100 \text{ km/h} \\ &= 27.78 \text{ m/s} \\ t &= 5 \text{ s} \end{aligned}$$

Acceleration

$$\begin{aligned} a &= (v - u)/t \\ a &= (27.78 - 0)/5 \\ a &= 5.56 \text{ m/s}^2 \end{aligned}$$

Force Required

$$\begin{aligned} F_{acc} &= ma \\ F_{acc} &= 1000 \times 5.56 \\ F_{acc} &= 5560 \text{ N} \end{aligned}$$

Distance travelled

$$\begin{aligned} s &= ut + (1/2)at^2 \\ s &= 69.5 \text{ m} \end{aligned}$$

Energy Required

$$\begin{aligned} E_{acc} &= F_{acc} \times s \\ E_{acc} &= 5560 \times 69.5 \\ E_{acc} &= 386.42 \text{ kJ} \\ E_{acc} &= 386.42/3600 \\ \mathbf{E_{acc} = 0.1073 \text{ kWh}} \end{aligned}$$

Equivalent Verification

Using kinetic energy

$$\begin{aligned} KE &= \frac{1}{2}mv^2 \\ &= \frac{1}{2} \times 1000 \times (27.78)^2 \\ &= 385.9 \text{ kJ} \\ &= \mathbf{0.1072 \text{ kWh}} \end{aligned}$$

Power Required

$$\begin{aligned} P &= E_{acc}/t \\ P &= 386.42/5 \\ P &= 77.3 \text{ kW} \end{aligned}$$

So a battery containing only 0.107 kWh could theoretically deliver 77.3 kW for 5 seconds

Battery discharge Rate (C-Rate)

$$\text{C-Rate: } 77.3/0.1072 = 721C$$

Not practical

Real EV battery is around 50 kWh

$$\text{C-Rate: } 77.3/50 = 1.5C$$

Torque Required

$$T = F_{acc} \times \text{Wheel Radius}$$

$$T = 5560 \times 0.3$$

$$\mathbf{T = 1668 \text{ Nm}}$$

Force and torque required to propel a car: Summary



Target: A 1000 kg sedan to reach from 0 to 100 kmph in 5 second

Parameters	Value
Vehicle mass (Assume)	1000 kg ✓
Final speed	100 km/h ✓
Time	5 s ✓
Acceleration	5.56 m/s ² ✓
Tractive force	5556 N ✓
Force by one person	100 N ✓
Number of person required to push	56 ✓
Wheel radius (Assume)	0.3 m ✓
Wheel torque	1668 N·m ✓

~ 2000 Nm

Powertrain



Powertrain generates and delivers power to a vehicle's wheels, including the prime mover, transmission, and driveline

Tractive force

Prime mover?

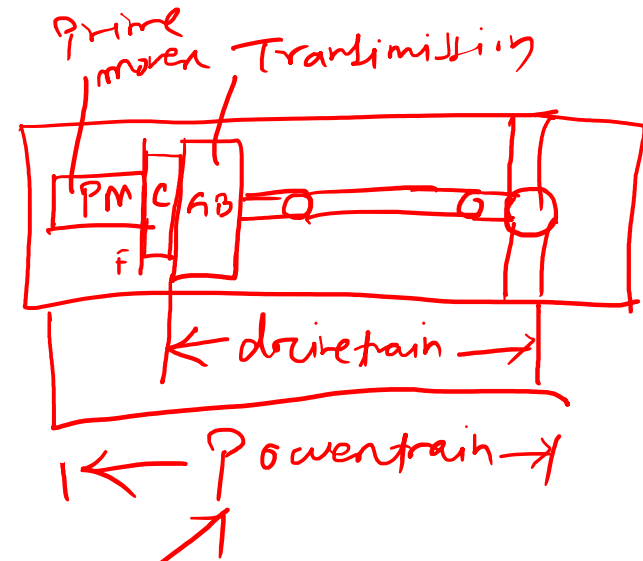
transmission?

Powertrain?

Drivetrain?

PM Excluded
AB → wheel

?



Powertrain Fundamentals



1. Vehicle Propulsion System Requirements



2. Powertrain Architecture



3. Tractive Force and Torque Requirement



What Is a Vehicle Propulsion System?

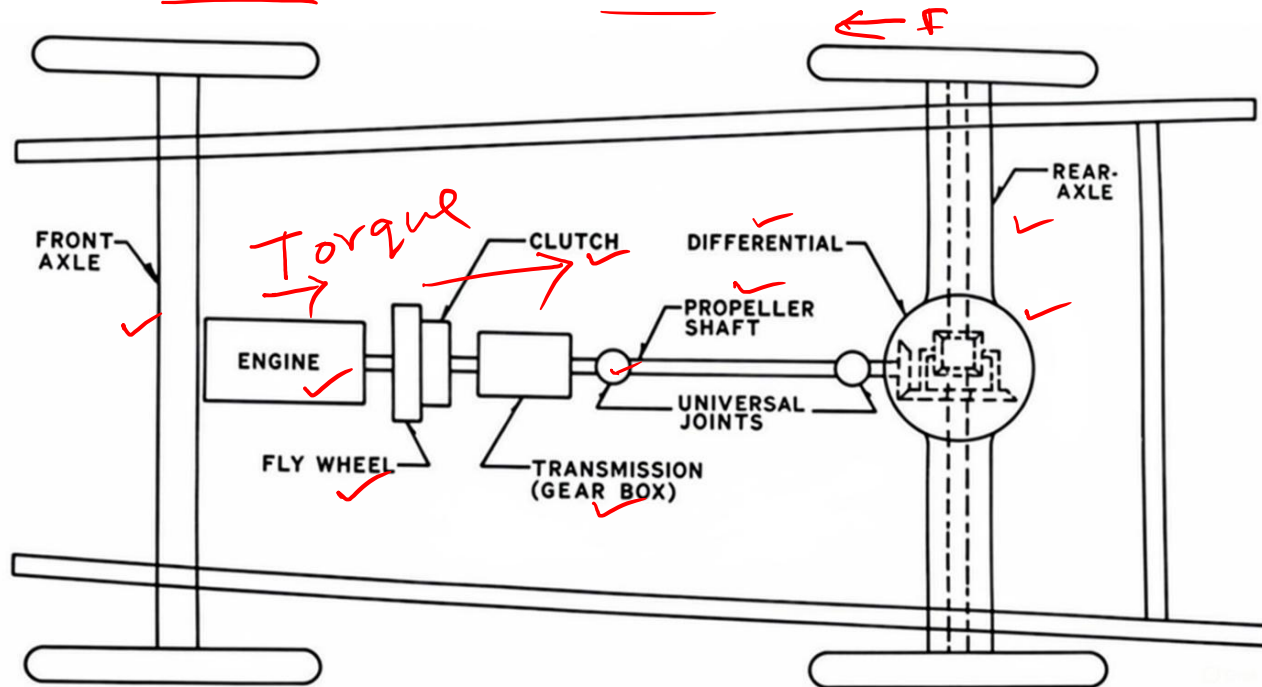


- The full path

Prime mover -> clutch or torque converter -> gearbox -> propeller shaft (if applicable) -> final drive/differential -> drive shafts -> wheels.

- Why each stage exists ?

Each stage exists to manage torque multiplication, speed matching, and the transfer of drive to wheels that must also steer and articulate over suspension travel.



Analogy between Human body and Automobile

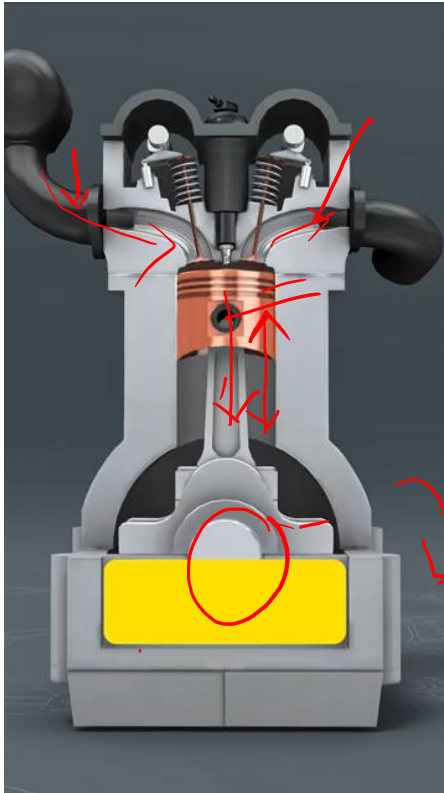


- Heart → Engine/Motor
- Lungs → Air intake system
- Kidney and liver → Oil filter and Exhaust
- Nervous system → Wiring harness
- Sweating → Radiator
- Muscles → Transmission and drivetrain
- Skeleton → Chassis
- Brain → ECU
- Blood → Fuel
- Skin → Body

Prime mover: Internal combustion engine



Prime mover: Converts chemical energy of fuel into mechanical power



❖ Design parameters:

- **Torque (Nm)** — T
- Power output (kW) — $WT = \frac{2\pi NT}{60}$
- **Fuel consumption (BSFC)** — $\frac{\dot{m}_f}{B.P} = \frac{o/p}{i/p} = \frac{B.P \rightarrow}{[Total] Energy Supplied}$
- Thermal efficiency (%) — $\eta_v = 100\%$
 $\eta_v = 90\%$
 $\eta_v = 50\%$
- Volumetric efficiency (%)
- Combustion efficiency (%)
- Engine size (cc or L)
- Emissions (HC, CO, Nox)

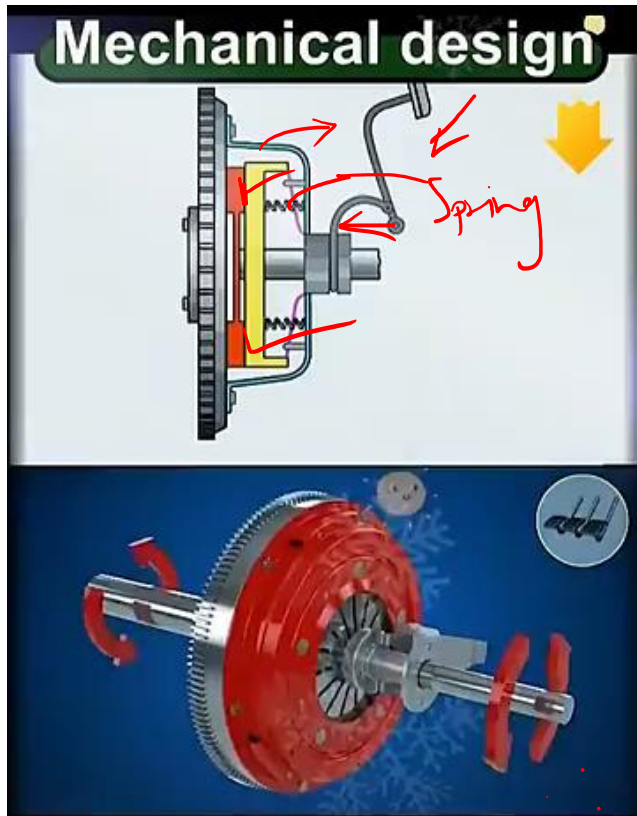


Working Principle of Spark-Ignition Engine

Clutch

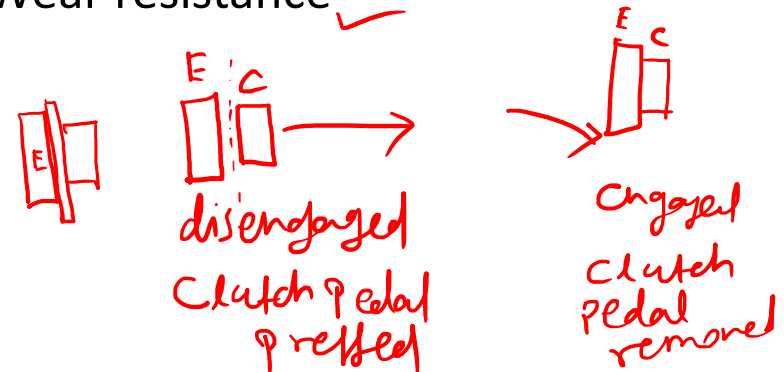
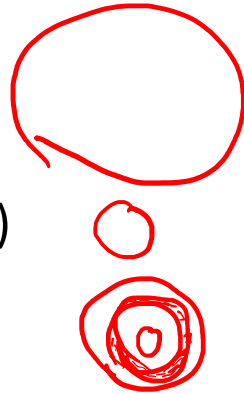


Prime mover: Connects and disconnects the engine from the transmission, smoothly.



❖ Design parameters:

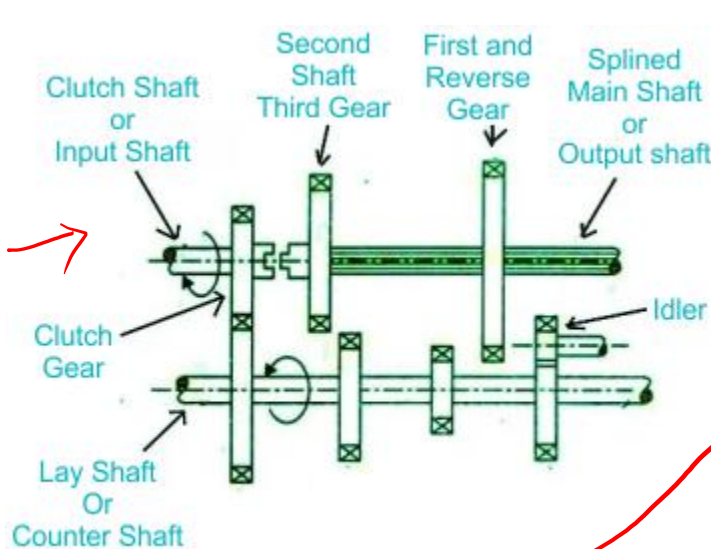
- Torque capacity (Nm) ✓
- Friction disc diameter (mm)
- Number of friction plates
- Clamp/spring force (N)
- Friction materials ✓
- Heat dissipation capacity
- Wear resistance ✓



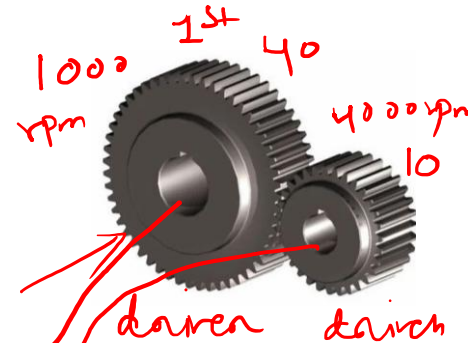
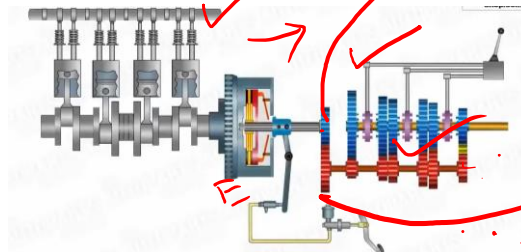
Working Principle of Clutch

Gearbox

Prime mover: Connects and disconnects the engine from the transmission, smoothly.



Sliding Mesh Gearbox



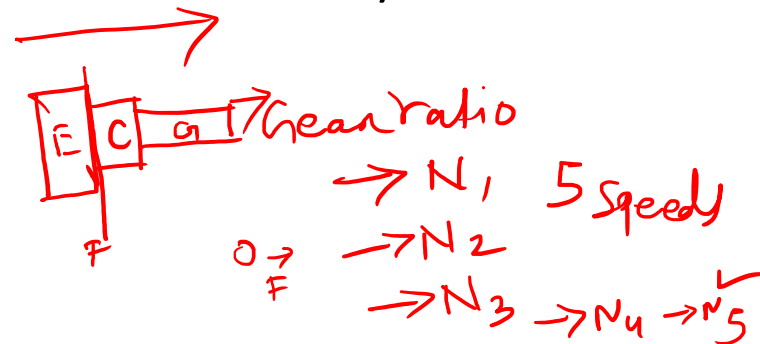
Gear Ratio

$$= \frac{\text{Teeth on Driven or output gear}}{\text{Teeth on Driver or input gear}}$$

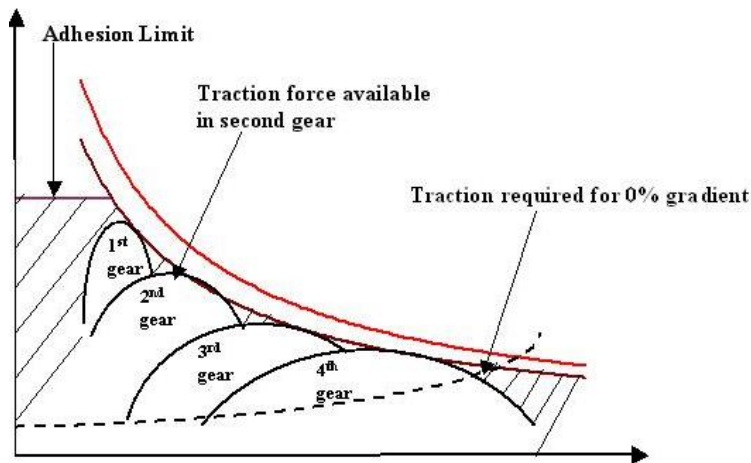
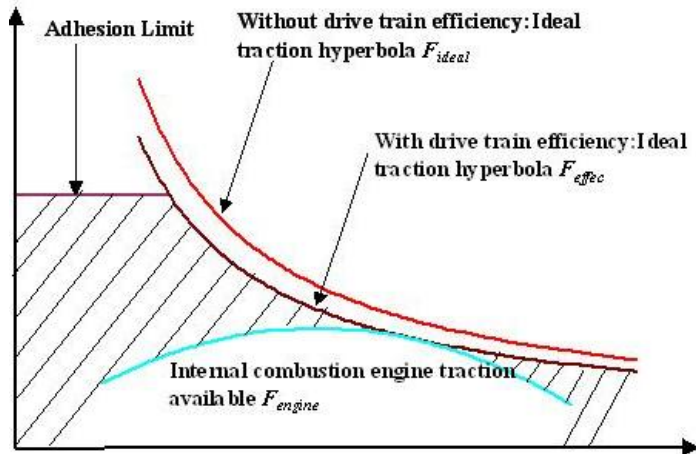
$$= \frac{T_{Dh}}{T_{Dr}} = \frac{40}{10} = 4 = \frac{N_{Dr}}{N_{Dh}}$$

❖ Design parameters:

- Torque transmission (Nm)
- Gear ratio (mm) ✓
- Gear efficiency



Gearbox: Why multi-speed



- ❖ **Tractive effort** is the driving force available at the wheels that allows a vehicle to accelerate, climb, and overcome resistances
- ❖ **Adhesion limit:** It is the maximum tractive (driving) force that can be transmitted between a vehicle's driven wheels and the road surface without wheel slip.
- When a wheel applies torque to the road, the force is transmitted through tire-road friction.
- This friction has a maximum value. Beyond this point, the tire loses grip and starts to slip or spin.
- That maximum usable force is called the adhesion limit.

Powertrain Fundamentals

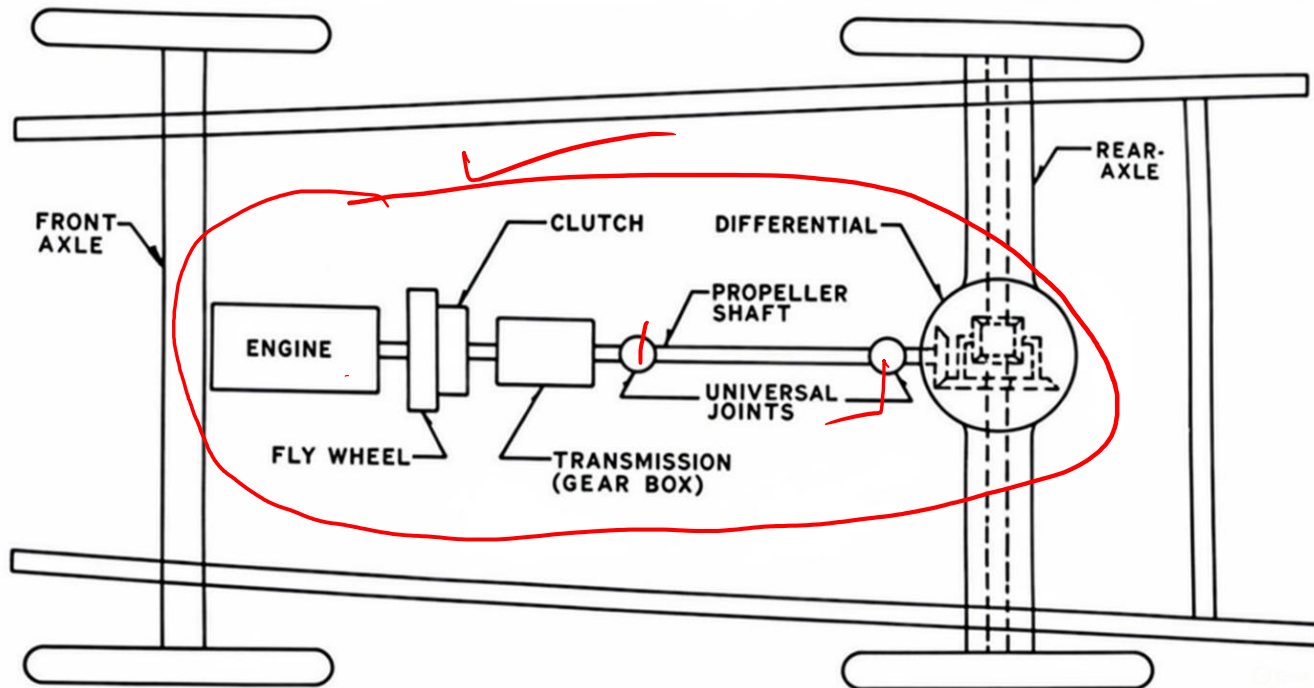


2. Powertrain Architecture

What Is Powertrain Architecture?



Powertrain architecture is the layout of the engine/motor, transmission, and driven wheels within the vehicle. *Universal arrangement? drive line*



Front and Rear wheel drives

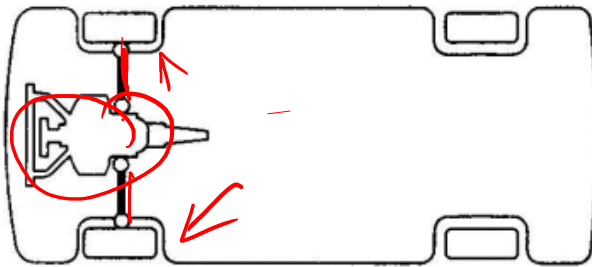
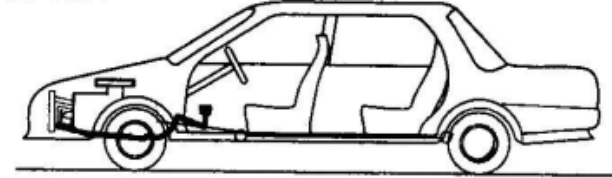
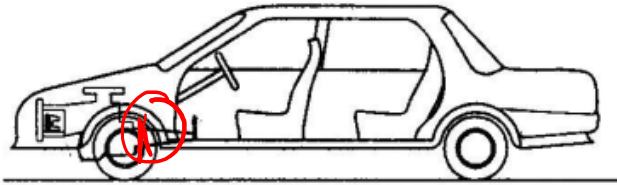


Fig. 1.22. Front-wheel drive.

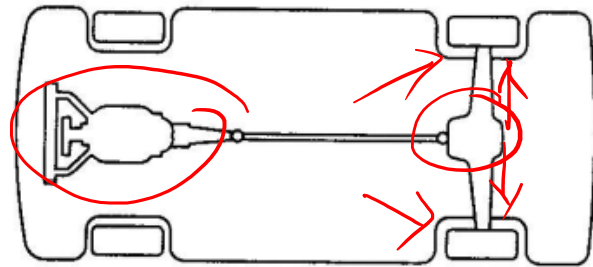
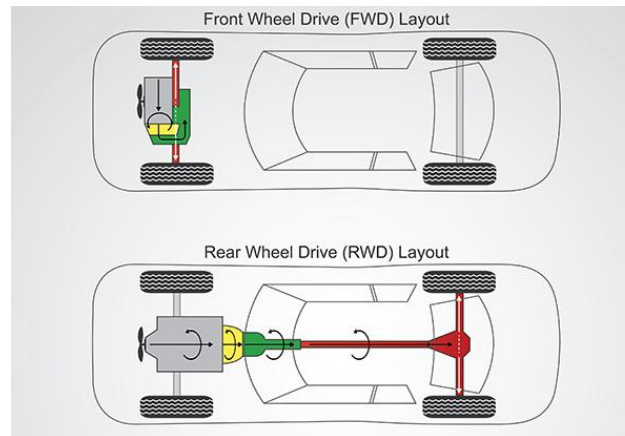


Fig. 1.21. Rear-wheel drive.



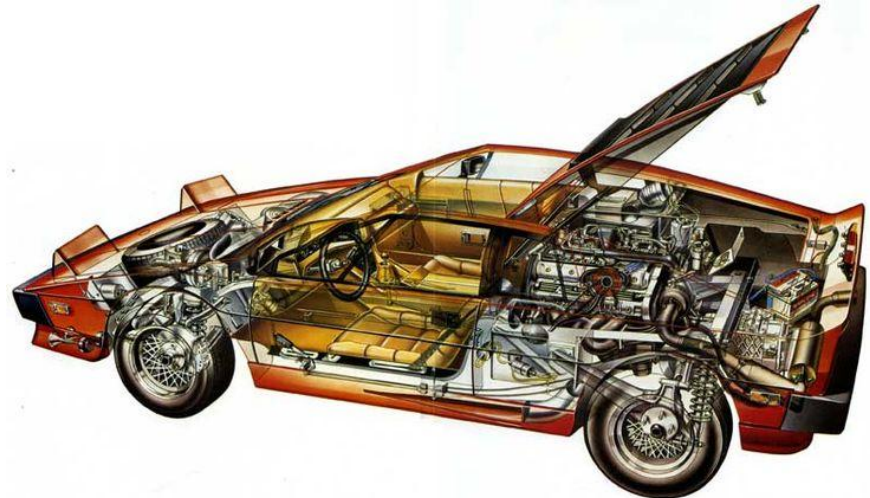
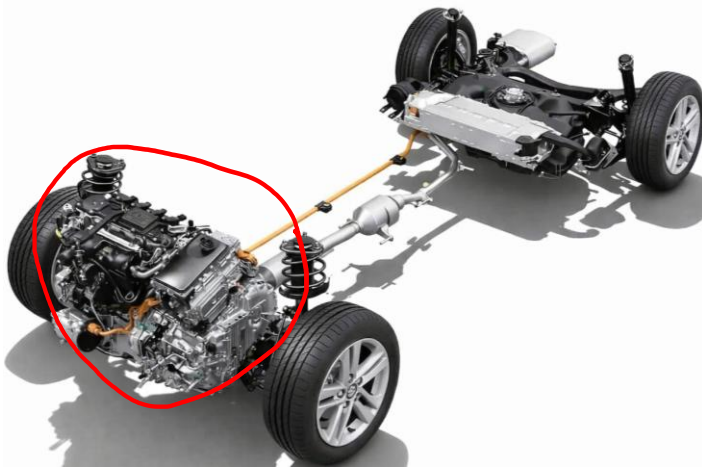
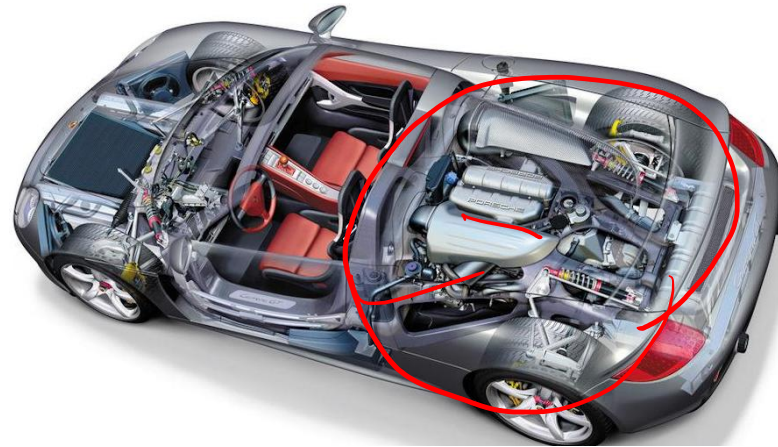
Front and Rear wheel drives



Front wheel drive



Rear wheel drive



Key Powertrain Layouts

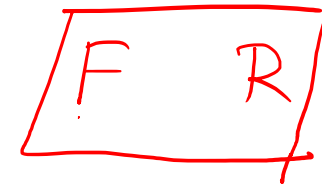


FF - Front-engine, Front-wheel-drive

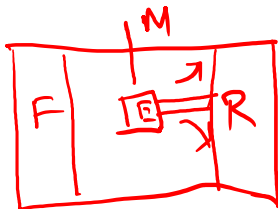
FR - Front-engine, Rear-wheel-drive

MR - Mid-engine, Rear-wheel-drive

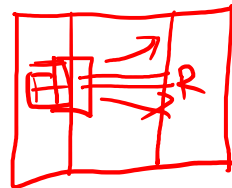
RR - Rear-engine, Rear-wheel-drive



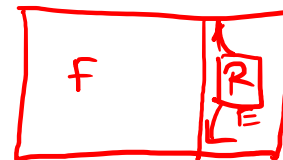
front engine
front wheel
drive



Mid engine
rear wheel
drive



Front Engine
rear wheel
drive



rear Engine
rear wheel
drive

